Tic-Tac-Toe with Unbeatable AI Using Minimax Algorithm

1. Abstract

This project implements a console-based Tic-Tac-Toe game in Python where a human player competes against an unbeatable Artificial Intelligence (AI). The AI uses the Minimax Algorithm to evaluate all possible future game states and select the best possible move.

2. Introduction

Tic-Tac-Toe is a simple two-player strategy game played on a 3×3 grid. The objective is to place three identical symbols in a horizontal, vertical, or diagonal row. In this project, the human player uses X and the AI uses O.

3. Objectives

• Develop a Tic-Tac-Toe game in Python.
• Implement AI using the Minimax Algorithm.
• Understand recursion and decision trees.
• Create an unbeatable AI opponent.

4. Tools and Technologies Used

Programming Language: Python
Algorithm Used: Minimax Algorithm
Platform: Console/Terminal
IDE: VS Code / PyCharm / IDLE

5. Working Principle

1. Display the game board.
2. Human enters a move.
3. Program checks for win/draw.
4. AI calculates the best move using Minimax.
5. AI places its symbol.
6. Process repeats until game ends.

6. Minimax Algorithm

The Minimax Algorithm is a recursive decision-making algorithm used in game theory. The AI tries to maximize its score while minimizing the human player’s score.

7. Main Functions Used

• print_board() – Displays the board.
• check_winner() – Checks winning conditions.
• is_draw() – Checks draw condition.
• minimax() – Implements AI logic.
• ai_move() – Finds best AI move.
• human_move() – Takes user input.
• play_game() – Controls game flow.

8. Advantages

• AI is unbeatable.
• Beginner-friendly implementation.
• Demonstrates practical AI concepts.
• Improves understanding of recursion.

9. Limitations

• Console-based interface only.
• Works only for 3×3 Tic-Tac-Toe.
• Minimax becomes slower for larger games.

10. Future Enhancements

• Add GUI using Tkinter or Pygame.
• Add difficulty levels.
• Implement Alpha-Beta Pruning.
• Add multiplayer support.

11. Conclusion

This project successfully demonstrates the implementation of Artificial Intelligence in a Tic-Tac-Toe game using the Minimax Algorithm. The AI always chooses the best possible move, making it impossible to defeat.

12. References

1. Python Documentation
2. GeeksforGeeks – Minimax Algorithm
3. IBM Artificial Intelligence Overview

Sample Code Snippet

```
def minimax(is_maximizing):
    if check_winner("O"):
        return 1
    if check_winner("X"):
        return -1
    if is_draw():
        return 0
```